Contents

The Decay Algebra: From Export Ledgers to a Continuum Entropy-Production Density

Paper C (decay algebra) — Draft Status: 2026-07-30 · Preliminary research · self-contained, **classical, no gravitational claims Scope contract:** ../DELIVERY_SCOPE.md · **Depends on:** Paper A ../01-foundations/PAPER_A_computational_entropy.md **Canonical notes:** `synthesis/m11f–m11i` · **Reproducible:** `simulations/classical/m11_{idem,decay_algebra,decay_algebra`

Abstract

Paper A defined computational entropy as the entropy of a map's output distribution and showed that the entropy exported by an irreversible computation, `H(X Y)`, is Landauer-exact but **path-dependent** and **non-additive**. That raises a sharp question: does a *coupled* lattice of local maps have a well-defined export **density** at all, and if so can it be computed without exponential enumeration? We answer both. First, the IDEM *decay vector* gives a tight upper bound on export, exact for maps with product (fully-recoverable) preimages. Second, promoting the hard decay flag to a **belief transported across the coupling boundary** yields a *decay algebra* — a transfer operator that computes the exact coupled export density. We prove a density theorem for the 1D nearest-neighbour case, generalize it to arbitrary gates and window widths (with a clean dichotomy: the algebra is exactly finite iff the gate has a fully-recoverable branch), lift it to a 2D strip transfer, and finally show a **hydrodynamic limit**: with a smoothly varying spatial field the discrete density converges (at rate `O(1/N)`) to a **constructive continuum entropy-production density field** `(x)`. Every result is backed by an exactly-reproducing computation. We make **no** claim that is a gravitational load term; that identification is discussed only as an explicit, deferred semantic bridge.

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1. Setup: from a path-dependent ledger to a density question

For a computation `X → Y` on uniform inputs, Paper A's **export identity** is

$$H(X) - H(Y) = H(X \mid Y) =: \text{export},$$

the chain-rule remainder (AND gate: `H(Y)=0.811278`, export `=1.188722`). Paper A also proved two facts that make a continuum density non-obvious: cumulative export `Σ L_S` is **path-dependent** (Thm 3), and the soft-recoverability slot is **non-additive** (Thm F).

So for a lattice of *coupled* local maps (maps sharing input wires), it is not clear that the total export is even extensive. We fix a concrete family: a sliding window of width `w` over an i.i.d. fair bit stream,

$$Y_i = g(b_i, \dots, b_{i+w-1}),$$

and study the per-site export density `a := lim_{k→∞} E(k)/k`, where `E(k)=H(X Y_{0:k-1})`.

Extensivity witness (`m11_lattice_export_density.py`). For coupled AND lattices `E(k)` is linear in `k` — a well-defined bulk density exists (shared-input `a 0.30076`, residual `~10 `; chained `a 0.999`) — and the path-dependence of Paper A saturates to an `O(1)` **boundary** term, so per-site it vanishes like `1/k`. The obstruction is real but subleading: a density exists.

2. The decay vector bounds export (exact for product maps)

The IDEM *decay vector* records, per input coordinate and per output branch, whether that input is recoverable from `Y` (`d_i=0`) or lost (`d_i=1`). Since `X (Y=y)` is uniform on the preimage `f ¹(y)`,

$$H(X \mid Y) = \sum_y p(y)\, \log_2 |f^{-1}(y)| \;\le\; \sum_y p(y)\, \#\{\text{unrecoverable coords}\} \;=\; \text{decay bound},$$

with **equality iff every preimage is a product subcube** (a fully-recoverable branch). This is proved and asserted over a map family (`m11_idem_export_density.py`):

map	exact export	decay bound	gap	product?
`erase(2,1)`, `erase(3,1)`, `erase(4,2)`	= bound	= exact	0	
`AND(2)`	1.18872	1.50000	0.311	
`majority(3)`, `parity(3)`	2.00000	3.00000	1.000	

Consequence (utility). For product/erasure lattices the export density is read directly from local decay metadata in `O(N)` — enumeration-free and exact. For non-product/coupled maps the bound is loose (e.g. the shared-input AND lattice has true density `0.30076   1.5`), which motivates the algebra of §3.

3. The decay algebra and the 1D density theorem

The looseness of the hard bound has two sources — fractional entropy of non-product preimages, and correlations from shared wires. Both are cured by replacing the `{0,1}` flag with a **belief over the shared boundary bit that is transported site-to-site** (a transfer operator).

Theorem 1 (1D density theorem; `m11f`). *For the 1D nearest-neighbour AND lattice `Y_i=b_i b_{i+1}`, the density*

$$a = \lim_k E(k)/k \;\text{ exists and }\; = 1 - h_Y, \qquad h_Y = \sum_{r\ge 0} \rho(r)\, h_2(p_1(r)),$$

with `E(k)=ak+b+o(1)`. The belief filter ` _i=Pr(b_i=1 Y_{<i})` resets to `1` on `Y_i=1`, so it collapses to a **run-length Markov chain** with ` _0=1`, ` _{r+1}=(1- _r)/(2- _r)`, `p_1(r)= _r/2`, stationary

law `(r) _{j<r}(1-p_1(j))`. Fixed point `=(3-√5)/2`, density

$$\boxed{a = 0.3007568}.$$

Proof idea. `(Y_i)` is a sliding-block factor of i.i.d. stationary ergodic the entropy rate exists (Shannon–McMillan–Breiman); the input rate is exactly `1`, giving `a=1-h_Y`. The filter is a sufficient statistic, and its reset structure makes the run-length chain ergodic with the stated stationary law. The belief-transfer recursion reproduces the `O(2^k)` enumeration to `<10` at `O(R)` cost (`m11_decay_algebra.py`).

4. General-`w` theorem: any gate, and the reset dichotomy

Theorem 2 (general `w`; `m11g`). *For any gate* `g:{0,1}^w→{0,1}` *and the sliding-window lattice:*

1. **(Existence.)** `(Y_i)` is `(w-1)`-dependent (outputs `w` apart use disjoint bits), hence stationary ergodic; `a=1-h_Y` exists with `E(k)=ak+O(1)`.
2. **(Blackwell representation.)** `h_Y = _{ }[H(Y )]`, where is the unique invariant measure of the belief-transfer operator on the `2^{w-1}`-state boundary simplex.
3. **(Regeneration.)** If `g` is a **reset gate** — some output value has a *singleton* preimage (a fully-recoverable `[0,…,0]` decay branch) — the filter restarts at the same point mass on every reset, so the inter-reset cycles are i.i.d. and `h_Y` is given by **renewal-reward**; the algebra is then exactly finite.

Corollary (dichotomy). The belief-transfer algebra is exactly finite `g` is a reset gate. This is executably verified (`m11_decay_algebra_general.py` asserts `exact singleton-preimage`):

gate	`w`	reset?	method	density `a`
AND	2,3,4	yes	exact	0.30076 / 0.56568 / 0.74503
OR	2,3	yes	exact	0.30076 / 0.56568
threshold 2	3	no	Monte Carlo	0.23734

Thus the *decay-vector structure of a single gate* — whether some branch fully recovers its inputs — controls whether the *composition algebra* is finitely exact. The reset direction is proved; the converse (non-reset non-atomic) is left conjectural.

5. 2D strip transfer

Lifting to a 2D lattice (2×2-plaquette AND, `Y_{ij}=b_{i,j} b_{i,j+1} b_{i+1,j} b_{i+1,j+1}`), the boundary between processed and unprocessed sites becomes a **row-cut of width `W`** and the transfer becomes an operator on the `2^W` cut configurations — the Bayesian-filter form of a statistical-mechanics strip transfer matrix.

Result (`m11h`). The 2D per-site density exists (finite-dependence on ^{2}; proved). The width-`W` density `a_W=(W-h_{\text{row}})/(W-1)` computed by the strip operator matches exact enumeration (`m11_decay_algebra_2d.py`):

W	2^W	a_enum	a_transfer	|Δ|
2	4	1.68412	1.68429	1.7e-4
3	8	1.19251	1.19329	7.8e-4
4	16	—	1.03098	—

`a_W` converges (1.684 → 1.193 → 1.031). The row-cut carries `2^W` states and `2^{W-1}` emission branches, so both enumeration and transfer cost grow exponentially in strip width. This exponential wall is generic in `d 2` and is exactly why a genuine limit — not more enumeration — is required.

6. Continuum embedding: a density field

Let the local statistics vary in space: `b_j Bernoulli( (x_j))`, `x_j=j/N`, for a smooth field ` :[0,1]→(0,1)` with the 1D nearest-neighbour AND coupling. The homogeneous density at bias is `a( )=h_2( )-h_Y( )`.

Result (local equilibrium; `m11i`). As `N→∞`, the coupled per-site density converges to the *pointwise* homogeneous density,

$$\boxed{\sigma(x) = a(\theta(x)) = h_2(\theta(x)) - h_Y(\theta(x))} \qquad \text{at rate } O(1/N).$$

The belief filter mixes geometrically (mixing length `O(1)` sites), so its law at site `i` equals the homogeneous stationary law at ` (x_i)` up to the bias variation over one mixing length, `O(1/N)`. Witness (`m11_continuum_embedding.py`, ` (x)=0.35+0.30x`, interior discrepancy `Δ(N)`):

N	40	80	160	320	640
Δ(N)	2.54e-3	1.72e-3	1.00e-3	5.34e-4	2.75e-4

Halving ratios `1.48 → 1.71 → 1.88 → 1.95 → 2`, and `Δ·N` approaches a constant, i.e. `Δ(N)=O(1/N)`. The discrete export ledger therefore admits a **constructive continuum entropy-production density field** ` (x)`, with total ^{1} `dx = 0.29937` for the sample field (profile ` = 0.446, 0.379, 0.301, 0.220, 0.146` at `x=0,¼,½,¾,1`).

7. Scope, rigor, and non-claims

Statement	Rigor
Coupled export is extensive; density exists	witnessed
Decay vector upper-bounds export, exact for product maps	**proved**
1D density theorem `a=1-h_Y=0.3007568` (Thm 1)	**proved**
General-`w` existence + Blackwell + reset-regeneration (Thm 2)	**proved** (non-atomicity converse conjectural)
2D density exists; width-`W` strip transfer	**proved** (existence); convergence numerical
Continuum density field ` (x)=a( (x))`, `O(1/N)`	witnessed (general theorem open)

Explicit non-claims. `(x)` is a **classical** entropy-production density for a specific lattice family. It is **not** identified with the gravitational load term `|dS_c/d |`, **not** continuum `L( ,g)`, and carries **no** gravitational content. A general hydrodynamic theorem (all gates, 2D) is open. The map from to any physical load term is, at most, a separate labeled semantic step and is not undertaken here.

Takeaway. From the Landauer-exact export ledger, promoting the decay vector to a transported belief yields a compositional *decay algebra* that computes coupled export densities exactly (for reset gates) and, in a hydrodynamic limit, a constructive continuum entropy-production density field. IDEM's decay vector thereby acquires a proved, load-bearing role: its fully-recoverable-branch structure is exactly what makes the algebra finitely exact.

Appendix R — Reproducibility

```
.venv/bin/python simulations/classical/m11_lattice_export_density.py    # extensivity; a 0.30076
.venv/bin/python simulations/classical/m11_idem_export_density.py       # decay bound, exact fo
.venv/bin/python simulations/classical/m11_decay_algebra.py             # 1D: a=0.3007568,  =(3
.venv/bin/python simulations/classical/m11_decay_algebra_general.py     # general w; exact reset
.venv/bin/python simulations/classical/m11_decay_algebra_2d.py          # 2D strip; a_W converg
.venv/bin/python simulations/classical/m11_continuum_embedding.py       #  (x)=a( (x)), O(1/N),
```

Quantity	Value	Source
AND export `H(X|Y)`	1.188722	Paper A / decay bound
1D density `a`	0.3007568 (`=(3−√5)/2`)	Thm 1
AND density `a_w`, `w=2,3,4`	0.30076 / 0.56568 / 0.74503	Thm 2
2D `a_W`, `W=2,3,4`	1.68429 / 1.19329 / 1.03098	§5
continuum total (sample field)	0.29937	§6

Changelog

Date	Entry
2026-07-30	Initial draft: extensivity → decay bound → 1D theorem → general-`w` → 2D strip → continuum density `(x)`; classical, no gravity; reproducibility appendix.